UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 008 · Problems on Taylor's and Maclaurin's series · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S008·12	Deck 008, slide 12 (rendered .pptx)
T008 00:07:38	AV-summary transcript 008, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b

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1. 1 · Taylor expansion about a point (a, b)
2. 2 · Maclaurin expansion about the origin
3. 3 · Approximate values from a Taylor polynomial
4. Past-paper questions

1 Taylor expansion about a point (a, b)

Definitions

Expansion "in powers of (x − a) and (y − b)". Taylor's series of $f(x, y)$ about the point (a, b).
S008·3 · T008 00:00:37

Up to n-th degree terms. Keep every term whose total power of $(x - a)$ and $(y - b)$ is at most n.
ADDED FOR CLARITY T008 00:06:10

Statements in the class material

> "$f(x,y) \approx f(a,b) + (x-a)f_x(a,b) + (y-b)f_y(a,b) + \frac{1}{2!}\left[(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\right]$" S008·14 · S008·19

Key points

1. Write the general series first, then find the derivatives, then substitute. That is the order the lecture follows. ESA ★★ T008 00:00:21 · T008 00:02:07

2. For $e^x \cos y$ at $(1, \frac{\pi}{4})$ every derivative is $\pm\frac{e}{\sqrt{2}}$; only the signs differ, so take $\frac{e}{\sqrt{2}}$ out as a common factor. ESA ★★★ T008 00:04:54 · T008 00:05:19
3. If the question asks for third degree, you need the third-order partials $f_{xxx}, f_{xxy}, f_{xyy}, f_{yyy}$ too. ESA ★★★ T008 00:06:10
4. Derivatives that vanish at the point drop whole terms: for $\sin(xy)$ at $(1, \frac{\pi}{2})$ both first partials are 0. ESA ★★ S008·15 · S008·16
5. Differentiate carefully with the product and chain rules; the slide's own f_{yy} for $xy^2 + y\cos(x-y)$ has a sign slip (see the exam traps). ESA ★★ ADDED FOR CLARITY S008·17 · S008·18

Exam traps

Exam trap. For $xy^2 + y\cos(x-y)$: $f_{yy} = 2x + 2\sin(x-y) - y\cos(x-y)$, so $f_{yy}(1,1) = 1$, not 3 as slide 008·18 prints. ADDED FOR CLARITY S008·17 · S008·18

Exam trap. Every value on slide 008·3 is $\pm\frac{e}{\sqrt{2}}$, not $\pm\frac{e}{2}$ as said at T008 00:03:47.
ADDED FOR CLARITY S008·3 · T008 00:03:47

2 Maclaurin expansion about the origin

Definitions

Up to three terms. Keep going until there are three non-zero terms: "make sure that you will get three non-zero terms". S008·5 · T008 00:07:38

Statements in the class material

> *"Taylor's series about* $(0,0)$ *is given by* $f(x,y) \approx f(0,0) + f_x(0,0)\,x + f_y(0,0)\,y + \frac{1}{2!}\left[f_{xx}(0,0)x^2 + 2f_{xy}(0,0)xy + f_{yy}(0,0)y^2\right] + \frac{1}{3!}\left[f_{xxx}(0,0)x^3 + 3f_{xxy}(0,0)x^2y + 3f_{xyy}(0,0)xy^2 + f_{yyy}(0,0)y^3\right] + \ldots$*"* S008·10

Key points

1. About $(0,0)$ the constant term is often 0 ($\log 1 = 0$, $\sin 0 = 0$), so the series starts at first degree. ESA ★★ T008 00:07:57
2. "Up to three terms" means three non-zero terms; the lecture still writes all second-order terms. ESA ★★ T008 00:07:38 · T008 00:10:55
3. Collect and simplify at the end: $2x + y - 2x^2 - 2xy - \frac{y^2}{2}$. ESA ★ T008 00:11:11
4. Check a two-variable answer with the one-variable series of the combination (E2.2, Figure G2.1). ESA ★★★ added for clarity S008·6 · S008·11

Why?

Why does $\log(2x + y + 1)$ give $(2x + y) - \frac{(2x+y)^2}{2} + \ldots$?
It is $\log(1 + t)$ with $t = 2x + y$, and $\log(1 + t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \ldots$; the two-variable Maclaurin series must agree with it.
added for clarity S008·6

Exam traps

Exam trap. For $e^x \log(1 + y)$: $f_{xy} = \frac{e^x}{1+y}$, so $f_{xy}(0,0) = 1$, not 0 as slide 008·12 lists.
added for clarity S008·12

Exam trap. "First six terms" of $e^x \log(1 + y)$ needs third order: $y + xy - \frac{y^2}{2} + \frac{x^2y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$. The slide stops at three. added for clarity S008·12 · S008·13

3 Approximate values from a Taylor polynomial

Definitions

Approximate value. The value of a truncated Taylor polynomial at a point near the expansion point. S008·7 · T008 00:11:47

Key points

1. Pick an expansion point near the target where f is easy: $(1, -1)$ for $(0.9, -1.2)$, $(1, 1)$ for $(1.1, 0.9)$. ESA ★★ T008 00:12:10 · CW U1 P26
2. "The more terms you take, the better the approximation": the first-order estimate of $\tan^{-1}(-1.08)$ is off by 0.012, the second-order one by 0.001. ESA ★★ T008 00:14:18
3. Substitute $h = x - a$ and $k = y - b$ with their signs: $x = 0.9$ gives $h = -0.1$, $y = -1.2$ gives $k = -0.2$. ESA ★★ T008 00:15:13 · T008 00:15:28

Why?

Why expand about (1, −1) to evaluate at (0.9, −1.2)?
At $(1, -1)$ every value is exact ($\tan^{-1}(-1) = -\frac{\pi}{4}$, simple derivatives), and $h = -0.1$, $k = -0.2$ are small, so the higher terms shrink quickly.
T008 00:12:10 · T008 00:12:48

Exam traps

Exam trap. With $x = 0.9$, $h = x - 1 = -0.1$ (negative). A sign slip here changes the answer by 0.1. (added for clarity) T008 00:15:13

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. A two-variable Taylor or Maclaurin expansion is asked in six of them, always a 6–7 mark Q1 part: Jul 2023 Q1.b and Dec 2023 Q1.b ($\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$, second order and third degree), May 2024 Q1.b ($e^x \cos y$ about $(1, \frac{\pi}{4})$, the same as slide 008·3), Jan 2024 set 1 Q1.b ($\log(1 + x - y)$), set 2 Q1.b ($e^{ax} \sin by$) and June 2024 Q1.c ($\sin(x + 2y)$). Four ask for third-degree terms. Jan 2024 sets 3 and 4 have no Taylor question. Approximate values (topic 3) were never asked in these papers; they appear in class and in CW problem 26.

1 · Taylor expansion about a point (a, b)

ESA May 2024 (7 marks) UE22MA141A Q1.b · solved in W1.1

Expand $f(x, y) = e^x \cos y$ in powers of $(x - 1)$ and $\left(y - \frac{\pi}{4}\right)$ using Taylor's series up to second degree terms.

ESA Jul 2023 (7 marks) UE22MA141A Q1.b · solved in W1.4

Expand $f(x, y) = \sin x \sin y$ in Taylor's series up to second order terms about the point $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

ESA Dec 2023 (7 marks) UE22MA141A Q1.b · solved in W1.5

Expand $f(x, y) = \sin x \sin y$ in Taylor's series up to third degree term about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

2 · Maclaurin expansion about the origin

ESA Jan 2024 (7 marks) UE23MA141A set 1, Q1.b · solved in W2.4

Expand $\log(1 + x - y)$ about origin up to third degree terms.

ESA Jan 2024 (7 marks) UE23MA141A set 2, Q1.b · solved in W2.5

Expand $e^{ax} \sin by$ about origin up to third degree terms.

ESA June 2024 (6 marks) UE23MA141A Q1.c · solved in W2.6

Expand $f(x, y) = \sin(x + 2y)$ in Taylor's series up to third degree terms about the point $(0, 0)$.